

Fault Diagnosis is yet another feature which an embedded system should incorporate lest it's a simple home appliance where a fault wouldn't cause heavy losses of life or property. Fault diagnosis is usually done on the core system (not on components) but can be done otherwise as well. It requires the system to build assertions and test them. If they fail, then a fault recovery sub-system can be activated. Let's take the example of a home management and security system: if the temperature sensors and infrared cameras don't detect any rise in temperature and the electricity regulation system doesn't report any problem but the smoke detector reports smoke being generated then it might fail (depending on the design) the assertion test for 'fire outbreak' and call for fault diagnosis for the smoke detection system.

Resource management

Resource Management is mostly a software level task. However, as we said at the beginning of this chapter – software is impotent without hardware support. The hardware must support features which enable better



Fault Isolation is important to ensure robustness of an embedded system

software to be built. If the hardware doesn't send enough information or sends it in a difficult-to-process format to software, the design won't be optimum. The hardware used in the system should be able to work over 'messages' and be able to send data to the main system in a similar format. Since polling is not a good way to utilise computing resources, it should be able to send interrupts and the main system should be able to handle them. Processors should be able to execute multiple jobs in a round-robin style by dividing time between jobs (multi-tasking) and be able to call the supervising program (the OS) at times. These hardware features open possibilities for higher quality software for the system.